

CrabTransformer Training Report

Scenario: REAL | Loss: MSE | Lag: 0 Years | Current Spawner: Yes

— Training Configuration —

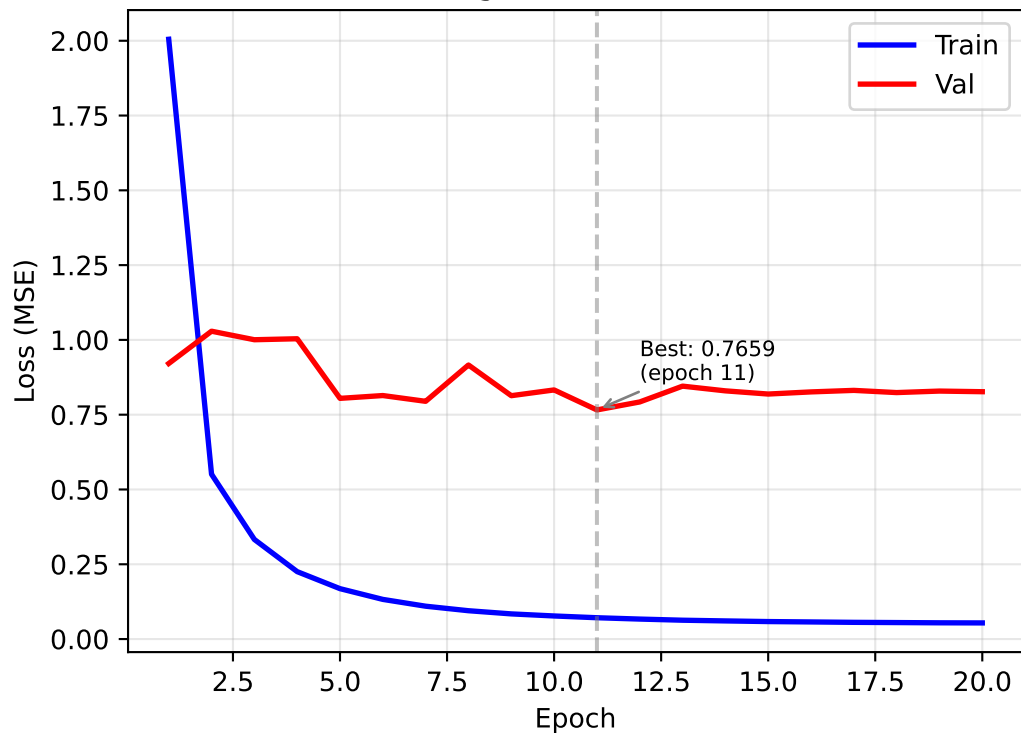
epochs_run: 20 best_val_epoch: 11
best_val_loss: 0.7659 final_train_loss: 0.0539
bias_correction (MSE): 1.0882
include_current_spawner: True

— Model Architecture —

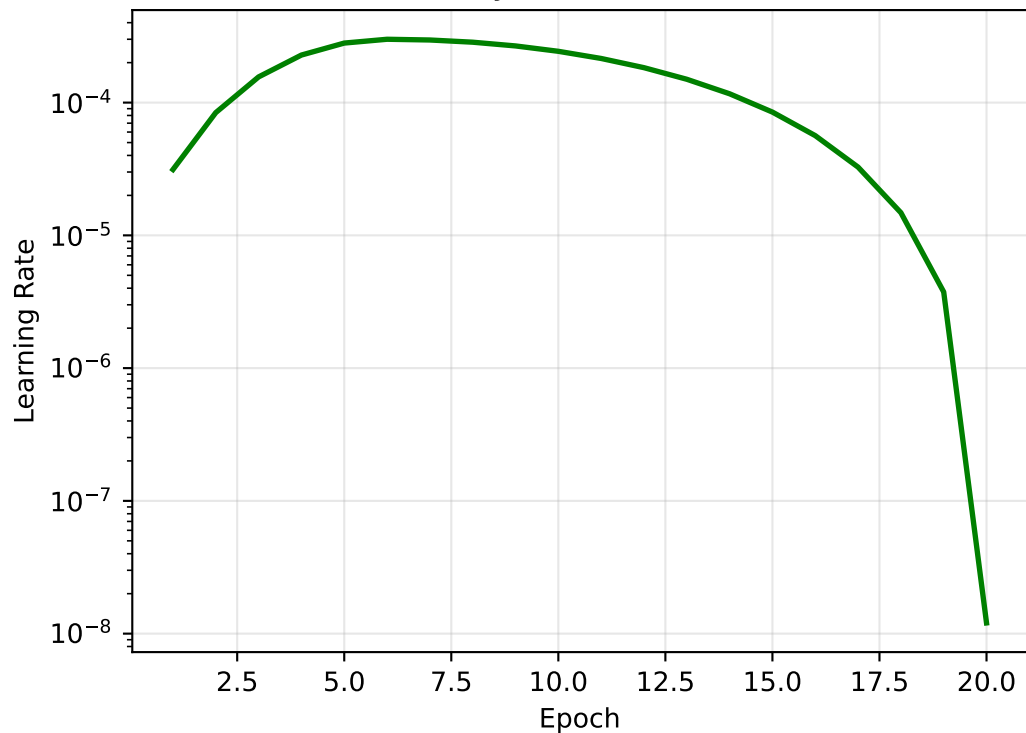
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 512 dropout: 0.2 patch_size: 5
train/val/test years: 24/8/4

Training Curves: REAL | Lag 0 | MSE | w/ Spawner(t)

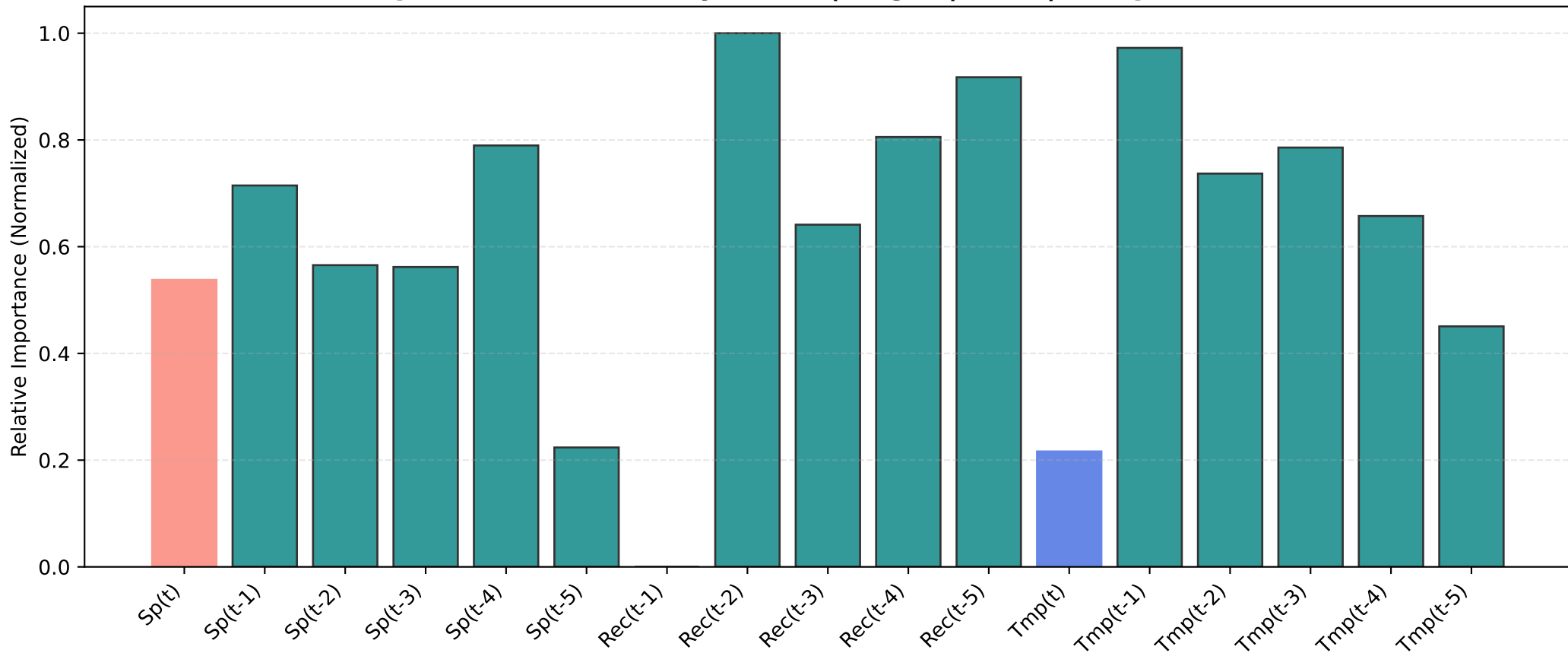
Training & Validation Loss



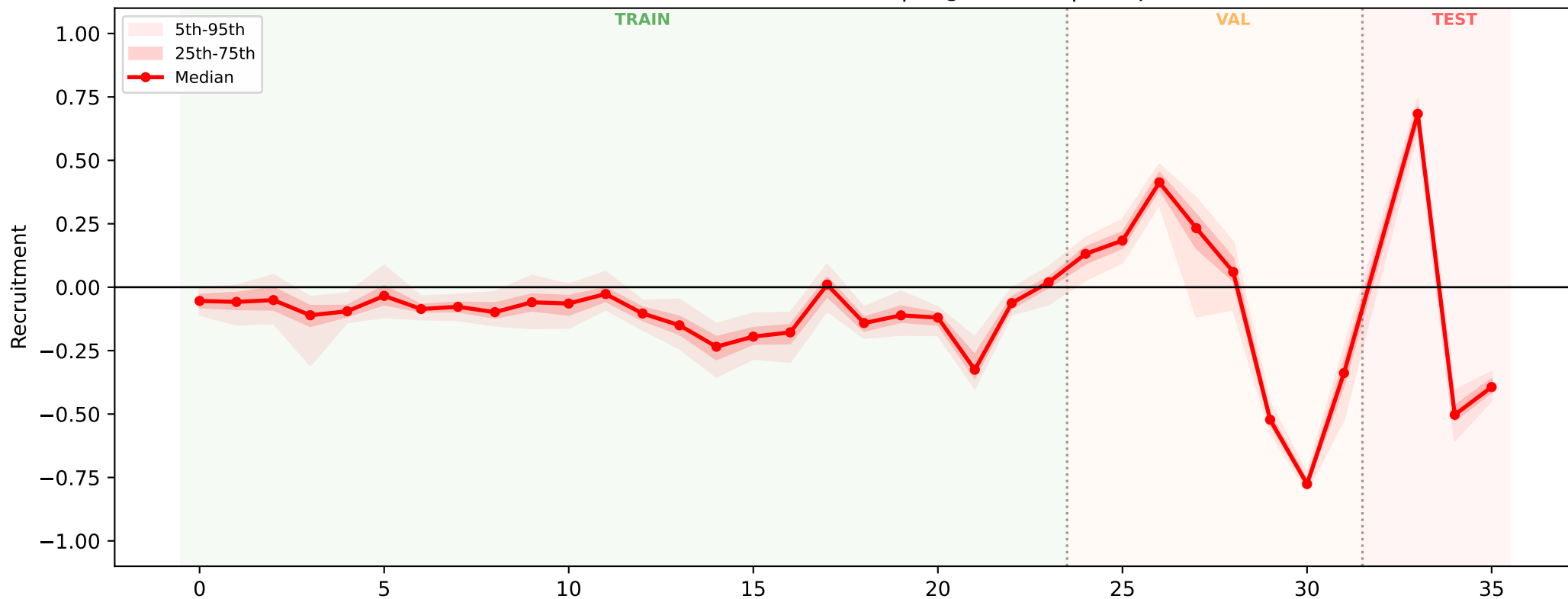
OneCycleLR Schedule



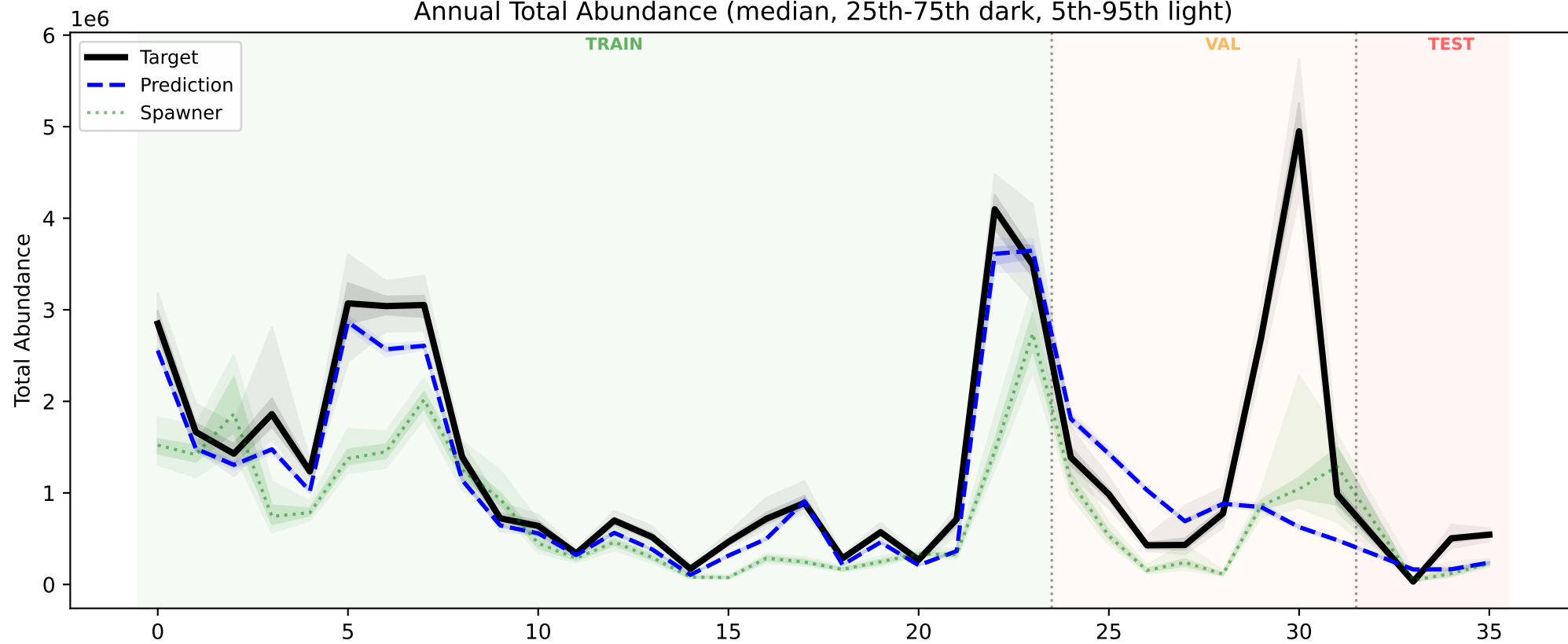
Input Channel Priority: REAL | Lag 0 | MSE | w/ Spawner(t)



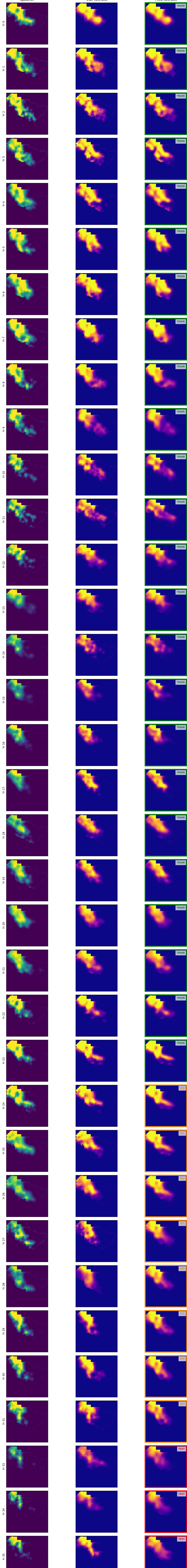
Annual Recruitment Deviations: REAL | Lag 0 (MSE) | w/ Spawner(t)



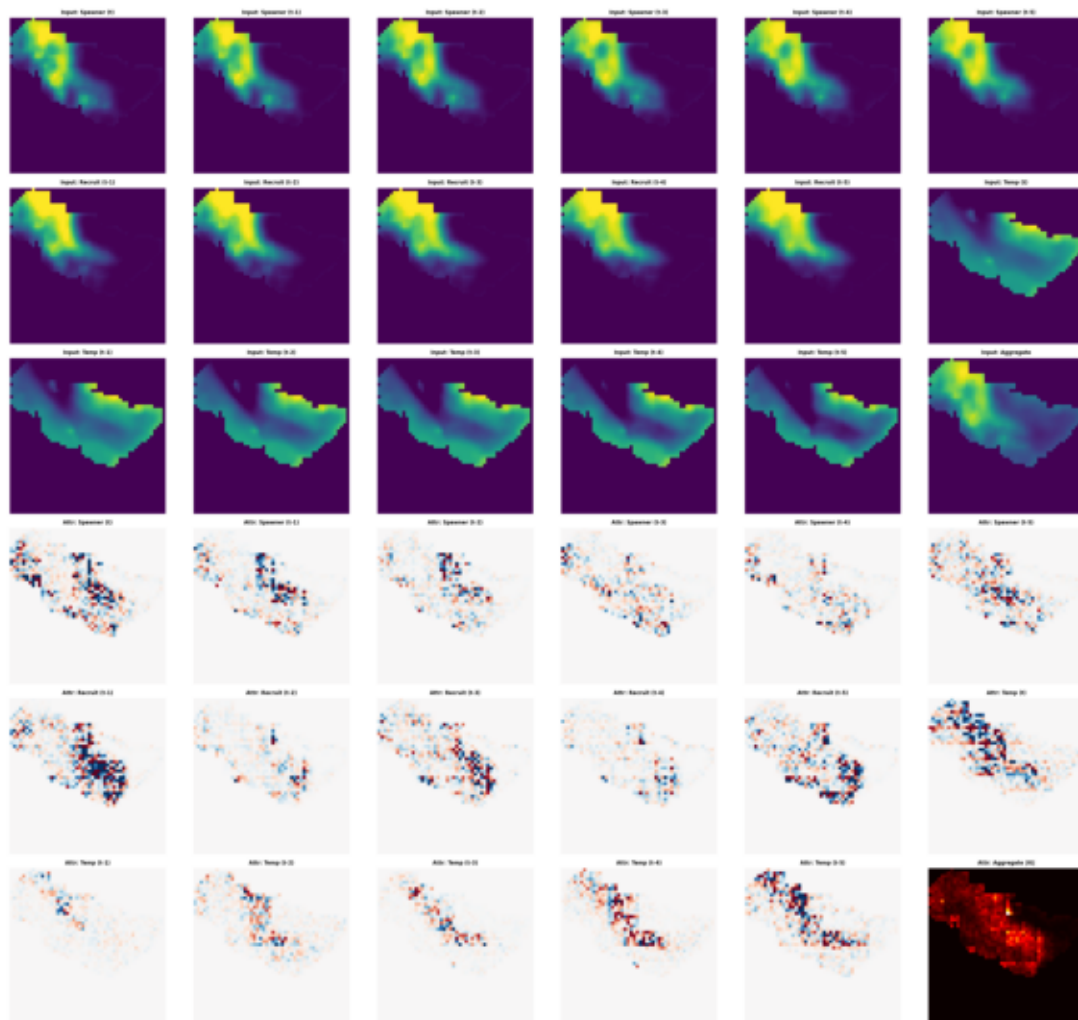
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



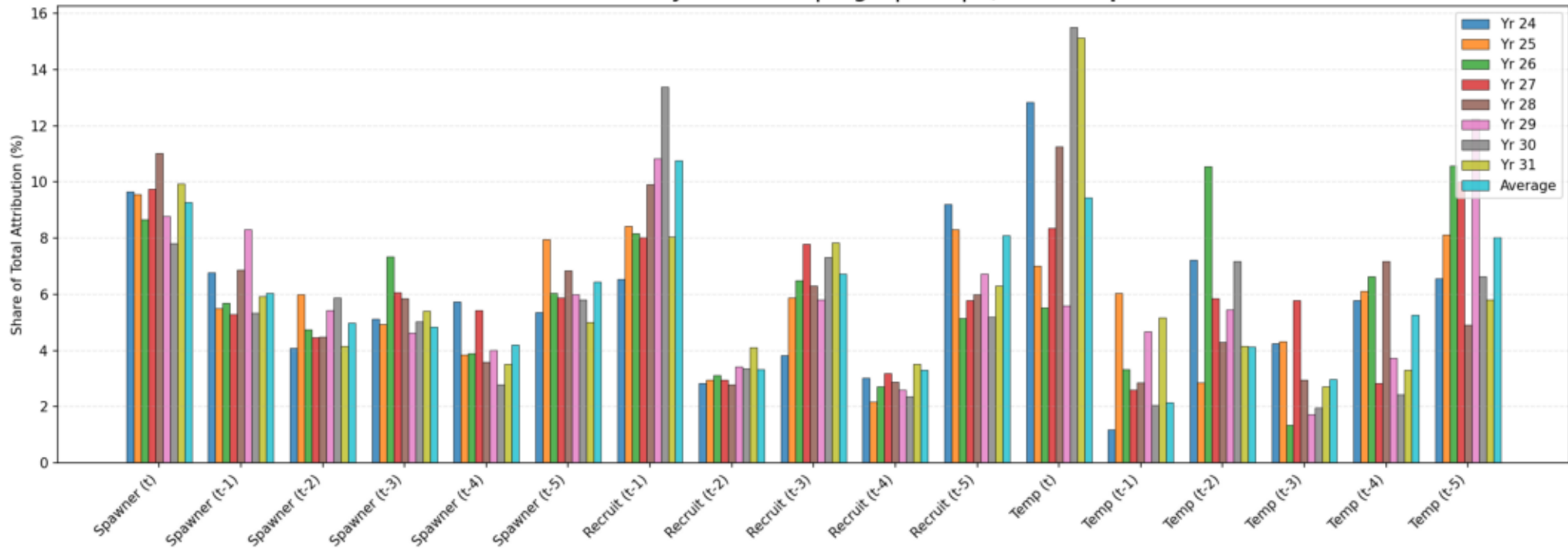
Spatial Trajectory: REAL | Lag 0 | MSE | w/ Spawner(t)



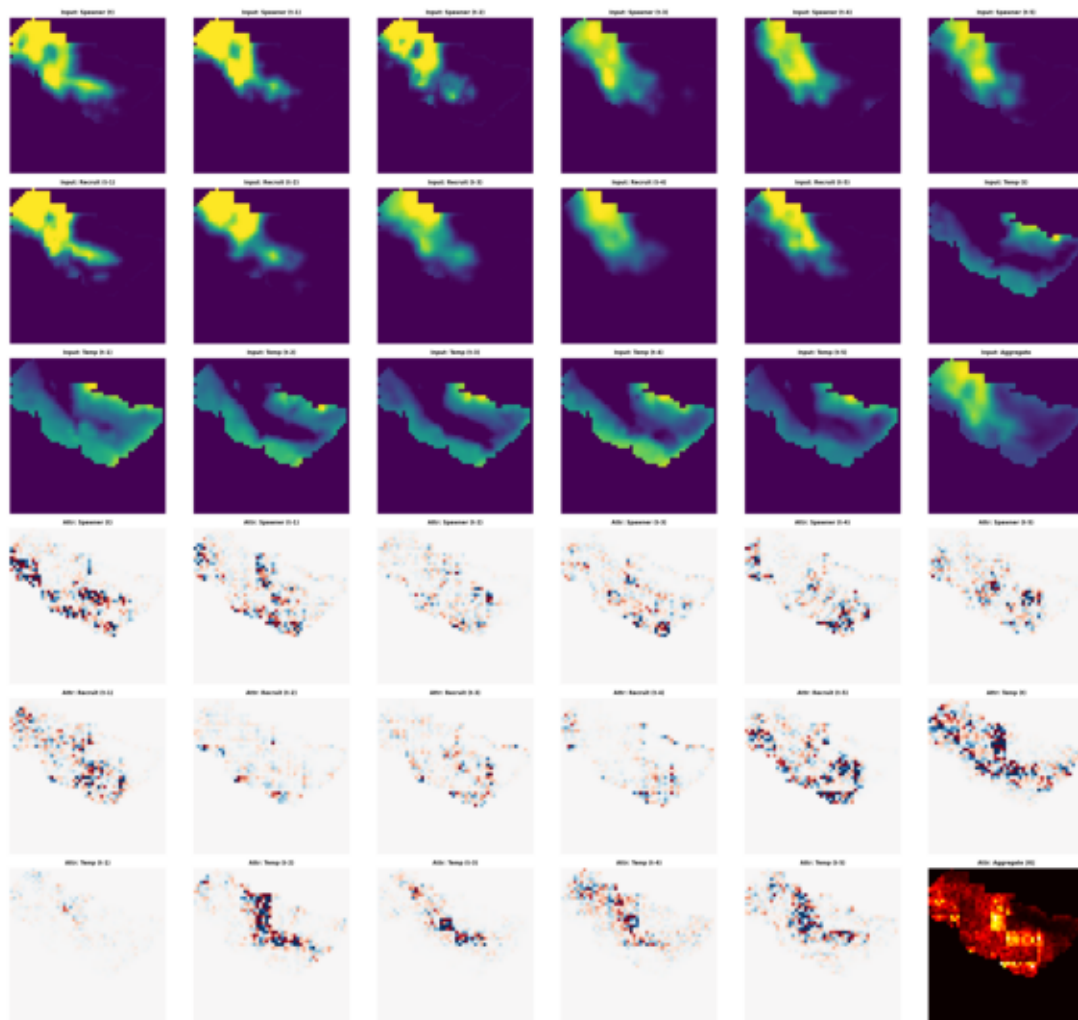
Year AVG -- REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



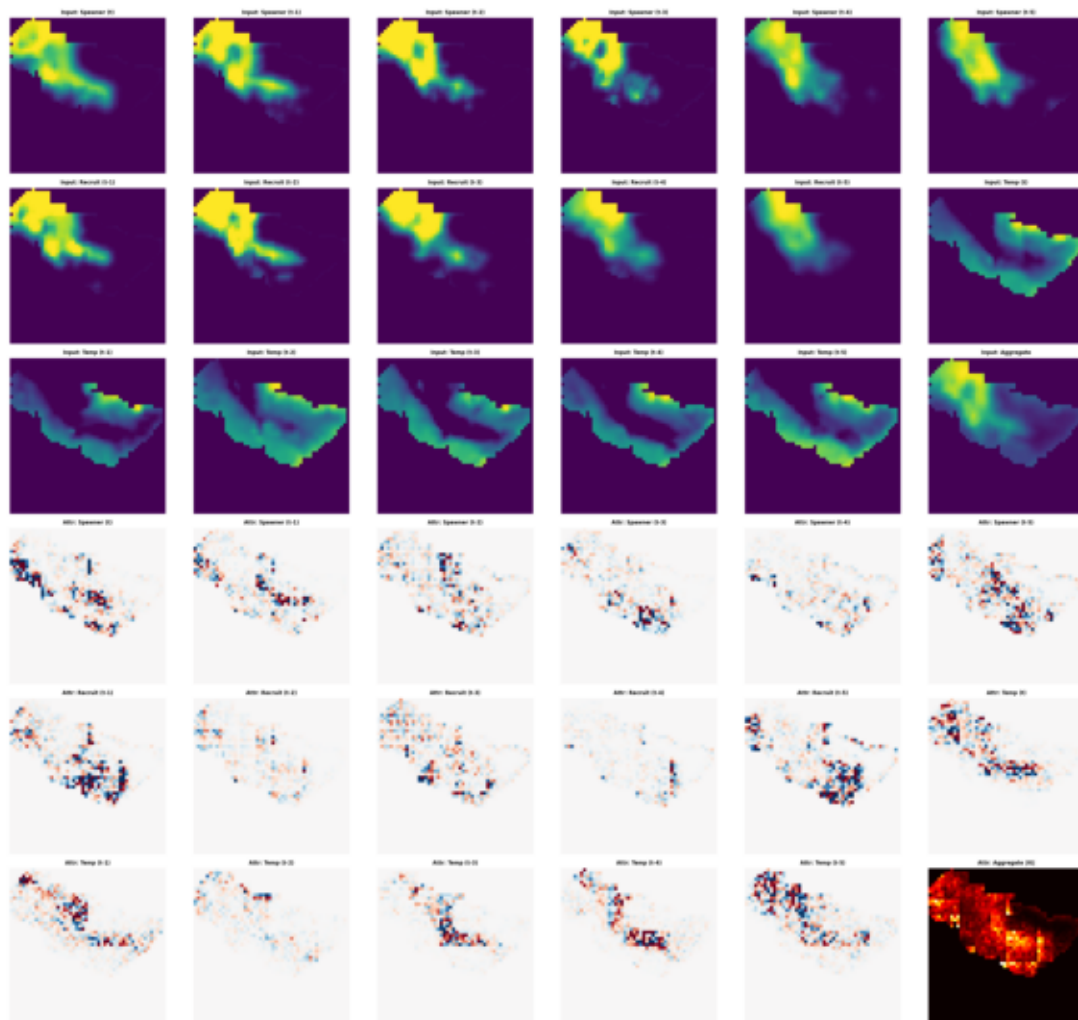
Channel Attribution by Year: REAL | Lag 0 | MSE | w/ Current Spawner



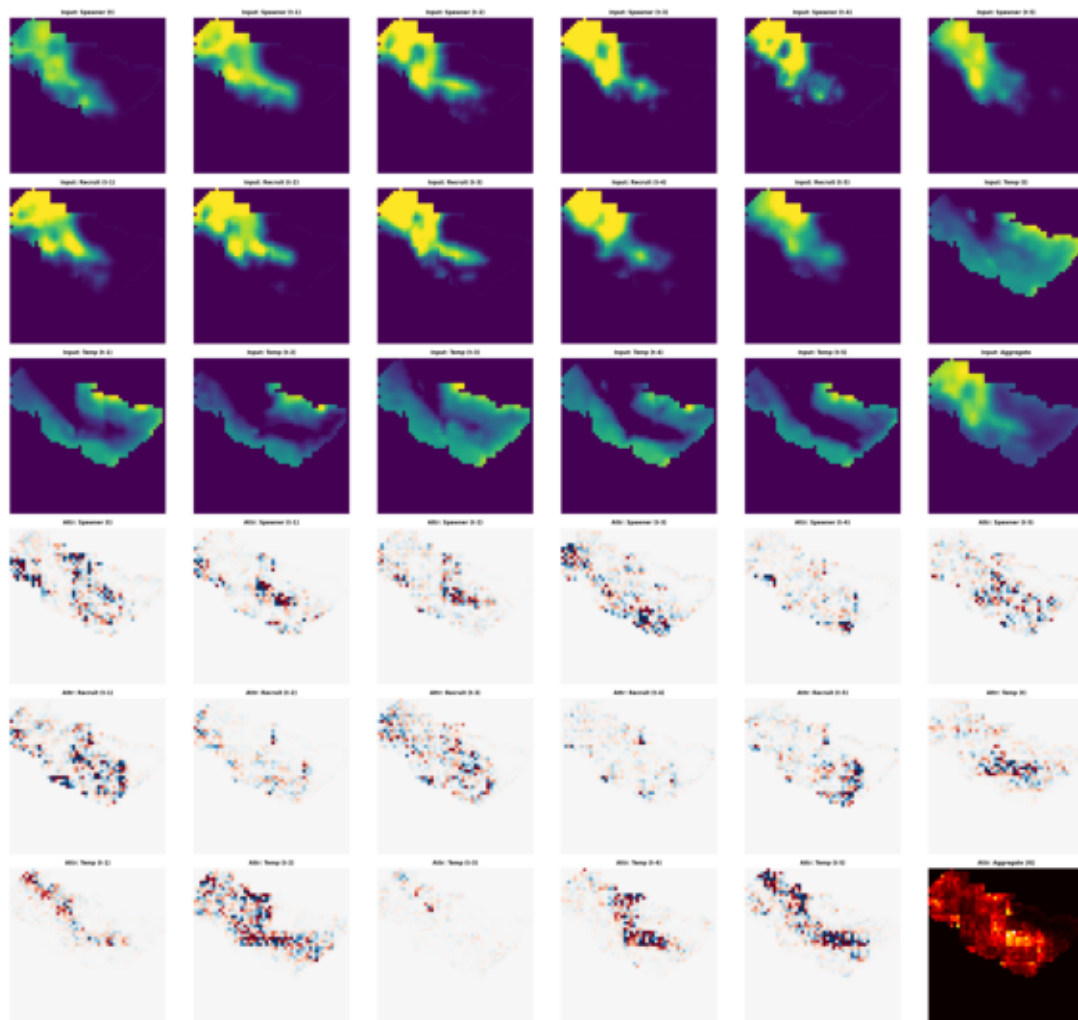
Year 24 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



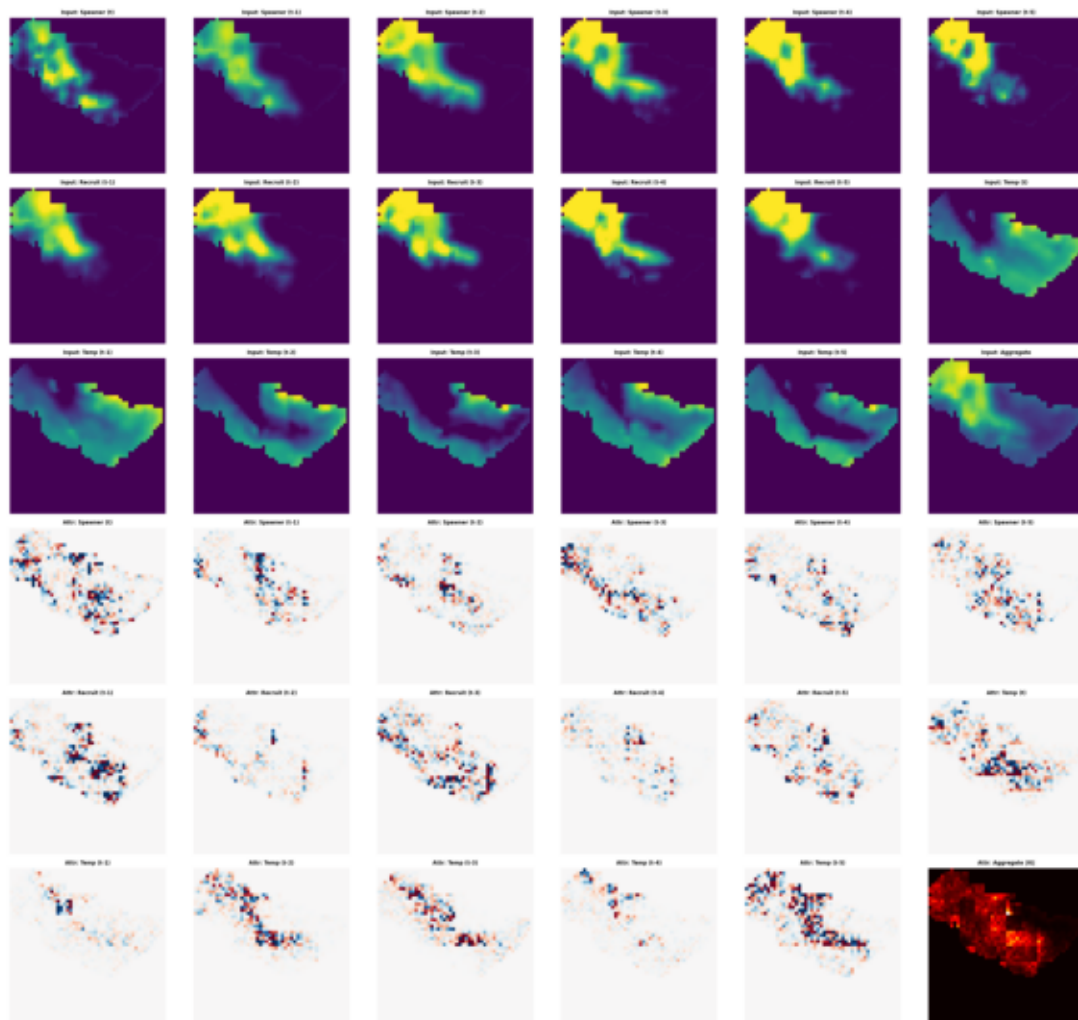
Year 25 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



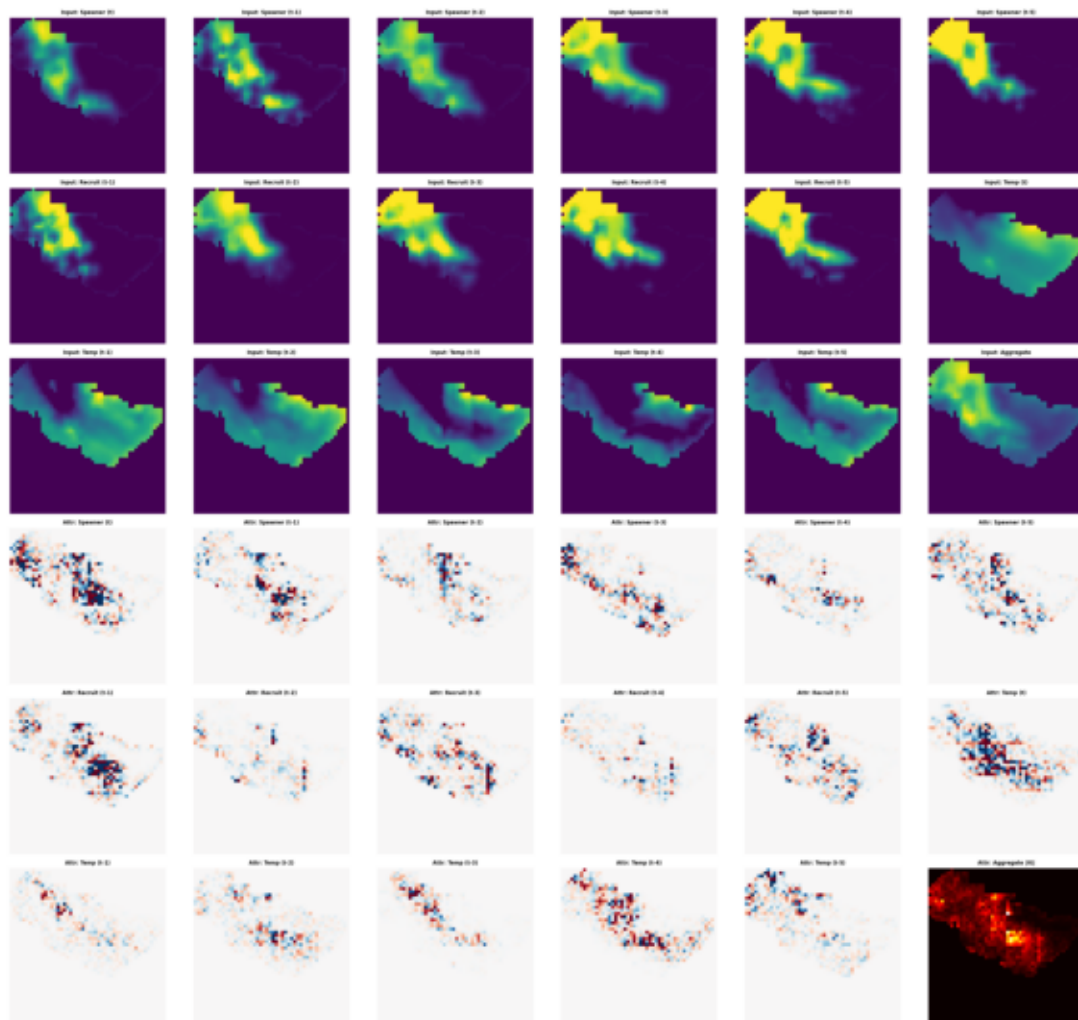
Year 26 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



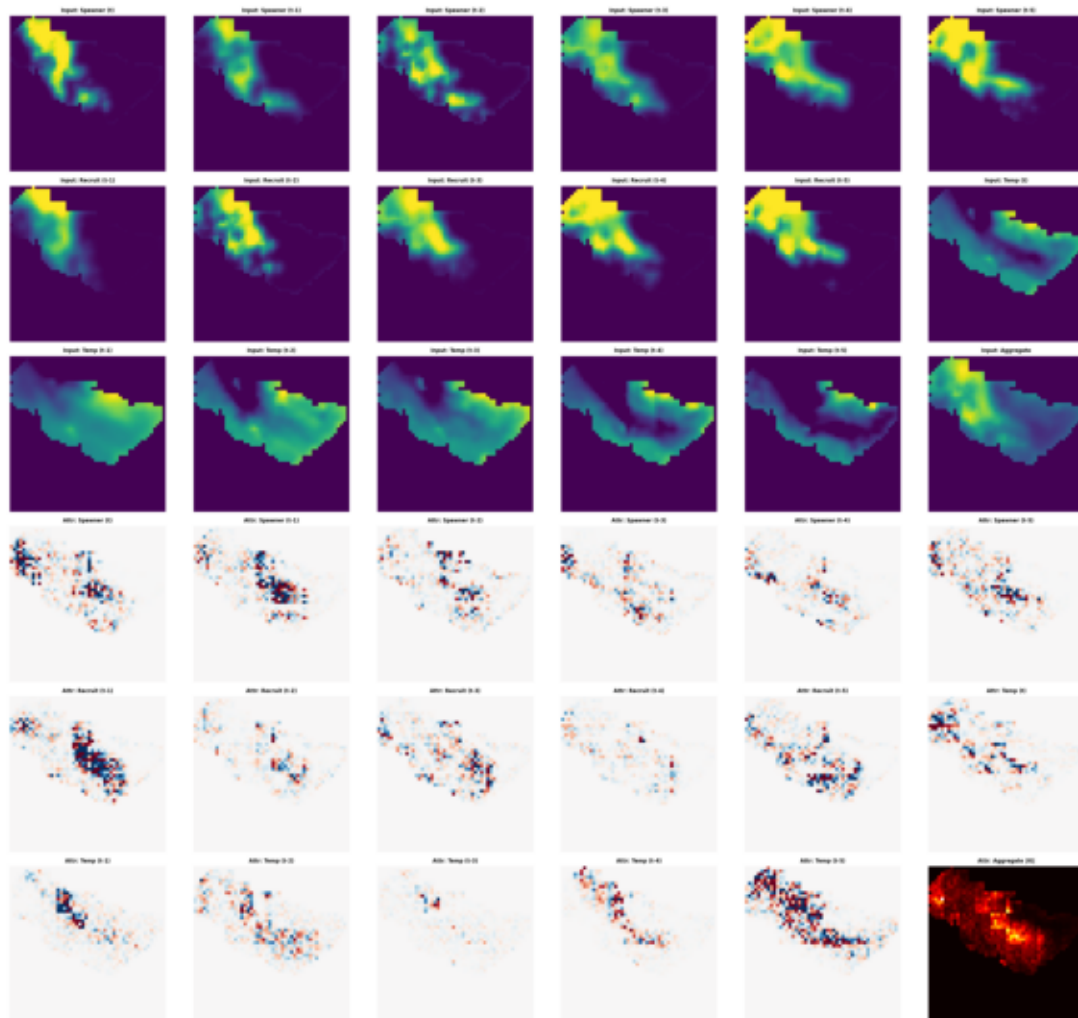
Year 27 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



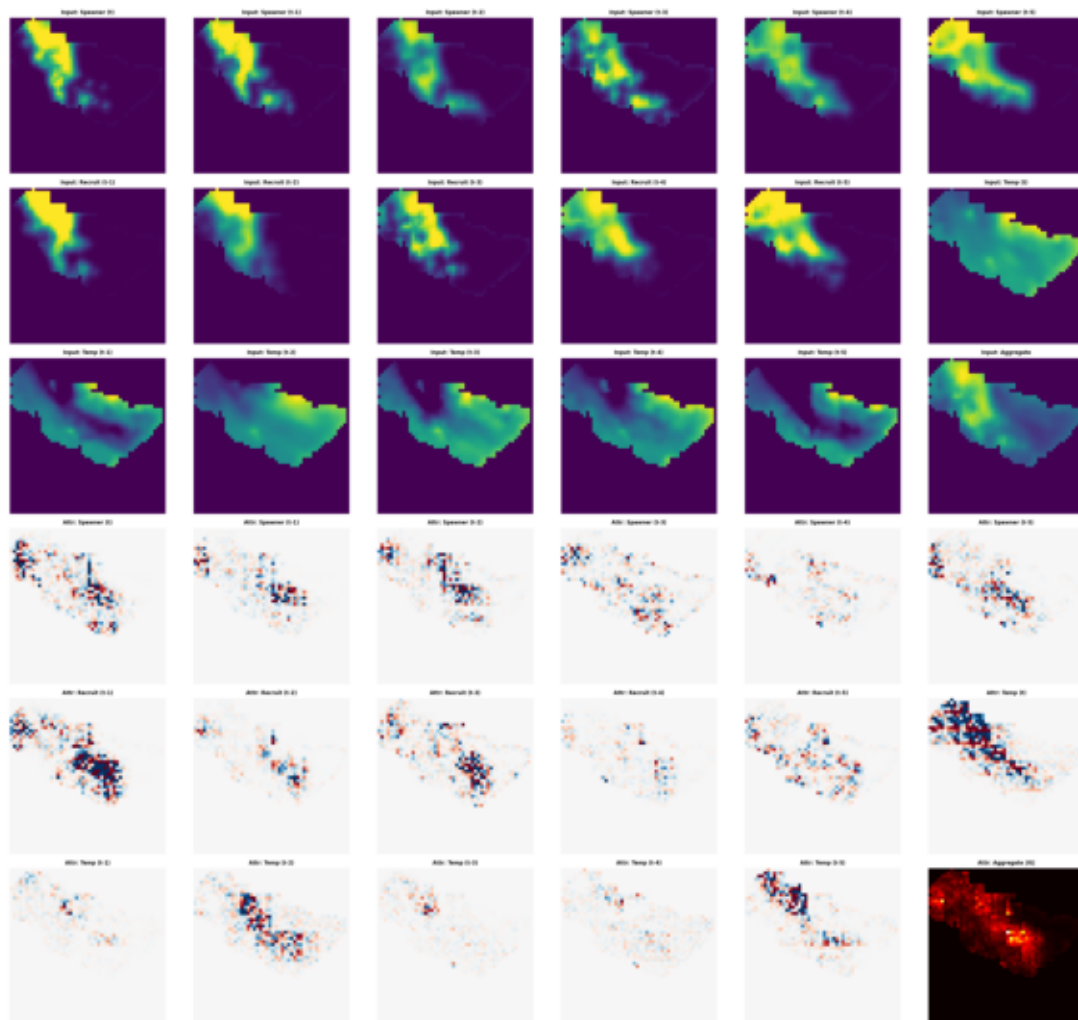
Year 28 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



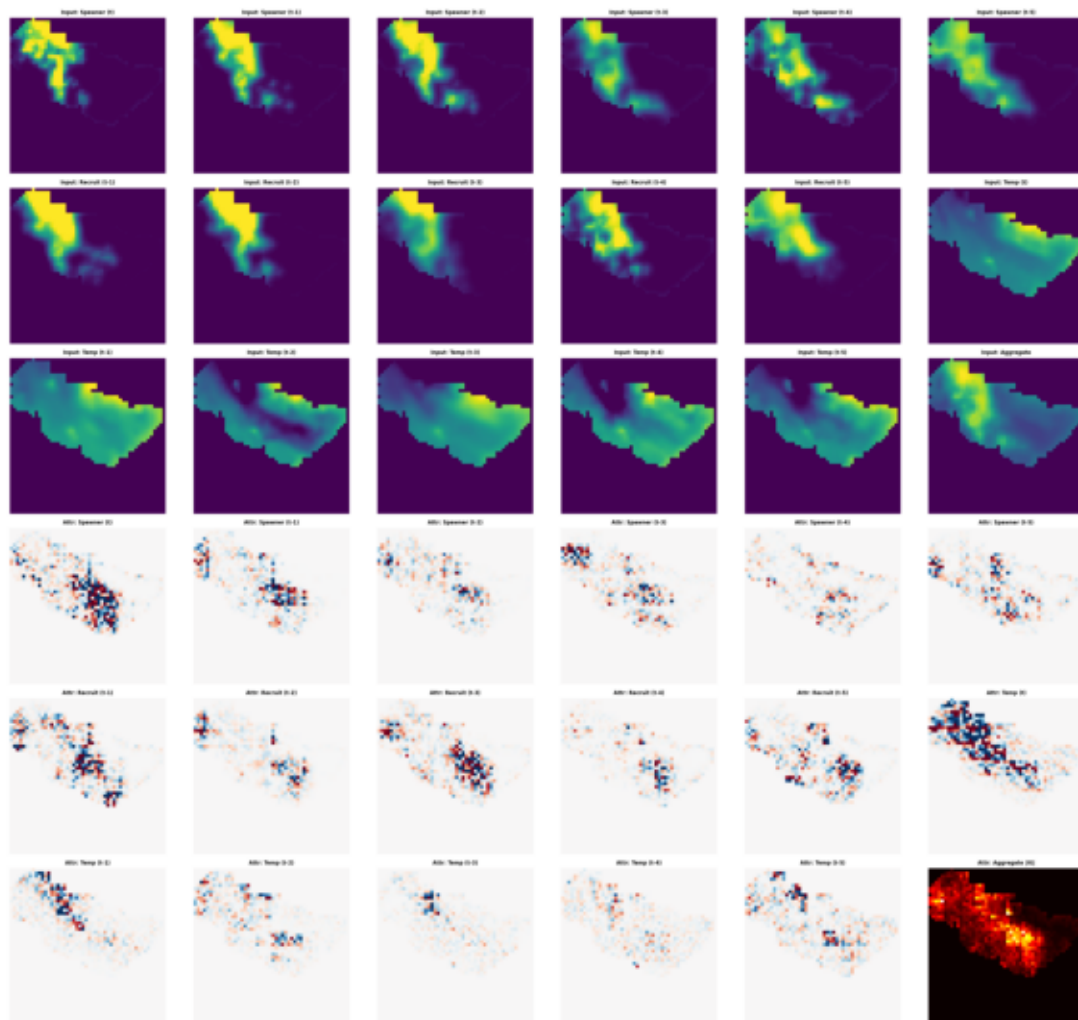
Year 29 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 — REAL | Lag 0 | MSE | w/ Current Spawner
Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 — REAL | Lag 0 | MSE | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 0 | MSE | w/ Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	1399.8	1612.8	5415.5	0.0	8182.8	0.868	455.7	0.974
VAL	1100.2	1801.5	3203.4	0.0	7582.9	0.611	1701.6	0.862
TEST	212.8	406.3	741.9	0.0	1316.3	0.524	335.9	0.849

Detailed Metrics (R-style): REAL | Lag 0 | MSE | w/ Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	1255.9	1231.1	-213.0	4.7%	0.983	0.976	-18.1%	0.983	0.927	0.865	0.17	0.905	0.841
VAL	5670.2	5530.4	-701.3	3.5%	0.866	0.620	7.4%	0.922	0.622	0.459	0.84	-0.444	0.312
TEST	1508.5	1476.5	-193.5	5.2%	0.863	0.726	106.4%	0.946	0.746	0.606	0.64	-15.037	-0.353